iOS
TIZEN
Operating Systems
Mac
Windows
Android
ios
Linux
Tizen
Debian

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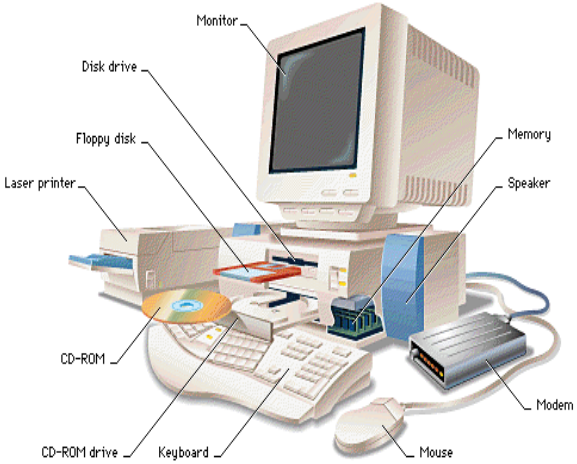
Objectives

1. What is an Operating System?
2. What does an OS do?
3. Structure of an OS
4. Evolution of Operating Systems
   - Batch Processing, Multiprogramming, Time sharing systems
5. Operating System Functions
6. Types of OS
   - Single User, Multi User systems

What is an OS?

- An Operating System is a piece of software, which makes a computer actually work.
- It is the software that enables all the programs that we use.
- The OS organises and controls the hardware.
- The OS acts as an interface between the application programs and the machine hardware.
- It hides the complexity of hardware from uninterested users.
- Examples: Windows, Linux, Unix, and Mac OS, etc.



What does an OS do?

An operating system performs basic tasks such as:

- Controlling and allocating memory
- Prioritising system requests
- Controlling input and output devices
- Facilitating networking
- Managing file systems.

Structure of an Operating System

- The structure of an OS consists of 4 layers:

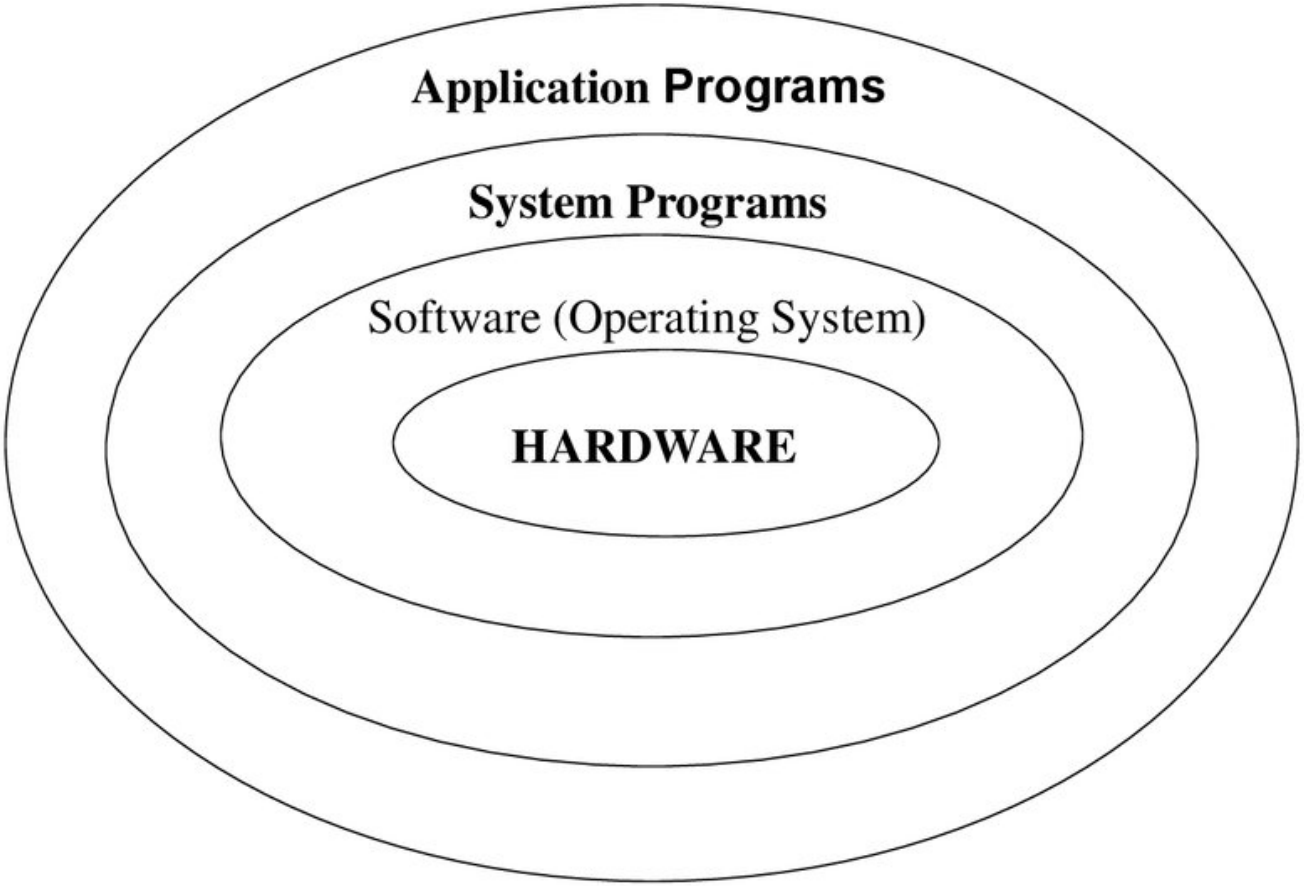
Hardware: Hardware consists of CPU, Main memory, I/O Devices, etc.

Software (Operating System): Software includes process management routines, memory management routines, I/O control routines, file management routines.

System programs: This layer consists of compilers, Assemblers, Linkers, etc.

Application programs: This is dependent on a user's needs e.g. reservation system, bank database management etc.

Structure of an Operating System



Evolution of OS

- The evolution of operating systems went through Five *major phases.*

- They have significantly changed the ways in which users accessed computers through:
  - Batch processing
  - Multiprogramming
  - Timesharing
  - Personal computing
  - Distributed systems

Evolution of OS

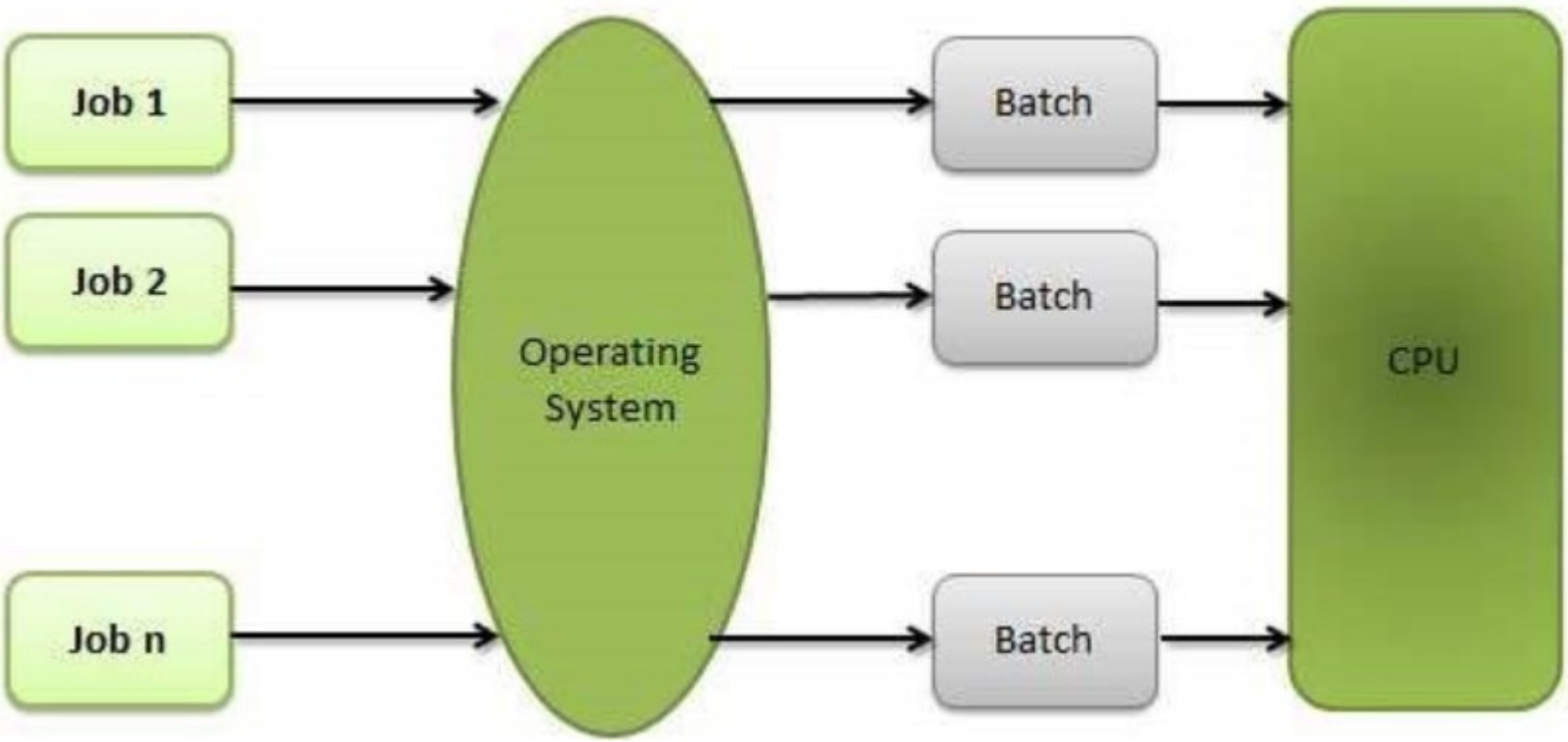
Major Phases	Technical Innovations	Operating Systems
Batch Processing	Tape batching, First-in, first-out scheduling.	BKS system (1961)
Multi- programming	Processor multiplexing, Indivisible operations, Demand paging, Input/output spooling, Priority scheduling, Remote job entry	Atlas supervisor (1961), Exec II system (1966)
Timesharing	Simultaneous user interaction, On-line file systems	Multics file system (1965), Unix (1974)
Concurrent Programming	Hierarchical systems, Extensible kernels, Parallel programming concepts, Secure parallel languages	RC 4000 system (1969), 13 Venus system (1972), 14 Boss 2 system (1975).
Personal Computing	Graphic User Interfaces	OS 6 (1972) Pilot system (1980)
Distributed Systems	Remote servers	WFS file server (1979) Unix United RPC (1982) 24 Amoeba system (1990)

Batch Processing

- Batch processing is a technique in which an Operating System collects the programs and data together in a batch before processing starts. An operating system does the following activities related to batch processing:

  - The OS defines a job which has a predefined sequence of commands, programs and data as a single unit.
  - The OS keeps a number of jobs in memory and executes them without any manual information.
  - Jobs are processed in the order of submission, i.e., first come first served fashion.
  - When a job completes its execution, its memory is released and the output for the job gets copied into an output spool for later printing or processing.

Batch Processing



Multiprogramming

- Multiprogramming is a technique to execute a number of programs simultaneously by a single processor.
- In Multiprogramming, a number of processes reside in main memory at a time.
- The OS picks and begins to execute one of the jobs in the main memory.
- If any I/O wait happened in a process, the CPU switches from that job to another job.
- **Advantages:**
  - Efficient memory utilisation
  - Throughput increases
  - CPU is never idle, so performance increases

Time Sharing Systems

- Time sharing, or multitasking, is a logical extension of multiprogramming.
- Multiple jobs are executed by switching the CPU between them.
- The CPU time is shared by different processes, so these types of systems are known as “Time sharing Systems”.
- The time slice is defined by the OS for sharing CPU time between processes.
- Examples: Multics, Unix, etc.

Operating Systems functions:

- The main functions of operating systems are:

  1. Program creation
  2. Program execution
  3. Input/Output operations
  4. Error detection
  5. Resource allocation
  6. Accounting
  7. Protection

Types of OS

Operating Systems can also be classified as:

- **Single User Systems**
- **Multi User Systems**

Single User Systems

- Provides a platform for only one user at a time.
- They are popularly associated with Desk Top operating system which run on standalone systems where no user accounts are required.
- For example, DOS.

Multi-User Systems

- Provides regulated access for a number of users by maintaining a database of known users.
- Refers to computer systems that support two or more simultaneous users.
- Another term for *multi-user* is *time sharing*.
- Example: All mainframes are multi-user systems.
  For example, Unix.

Introduction to Batch files

- In Windows, the batch file is a file that stores commands in a serial order. The command line interpreter takes the file as an input and executes it in the same order.

- A batch file is simply a text file saved with the .bat file extension. It can be written using Notepad or any other text editor.